Practical no. 7

Aim – Interface RFID module with Raspberry Pi to read and write to RFID tags/cards.

Theory –
RFID (Radio Frequency Identification) is a wireless technology used to identify and track objects using radio waves. It allows data to be read from a tag without physical contact or direct line of sight. The RFID reader continuously emits a 13.56 MHz electromagnetic field.
RFID is widely used in:

- Attendance systems
- Access control
- Library management
- Inventory tracking
- Toll collection
- Smart parking systems

RFID Tag -

- Contains a small microchip and an antenna.
- Stores a unique identification number (UID).
- Can be passive, active, or semi-passive.

SPI Communication (protocol)
SPI (Serial Peripheral Interface) is a synchronous communication protocol used between the Raspberry Pi and the MFRC522 module.
The SPI lines are:

Signal	Function
MOSI	Master Out Slave In
MISO	Master In Slave Out
SCLK	Serial Clock
SDA (CS)	Chip Select
VCC	Power Supply (3.3 V)
GND	Ground

- **Total Memory: 1 KB (1024 bytes)**
- **Sectors:** 16
- **Blocks per Sector:** 4

- **Block Size:** 16 bytes
- **Sector Trailer:** The last block in each sector stores **Key A, Key B, and access control bits** for security.

Memory Layout:

- Total Memory = **1024 bytes**

Types of RFID Tag Memory

1. **Read-Only (RO):** Data is written only by the manufacturer and cannot be changed.
2. **Write Once Read Many (WORM):** Data can be written once and read multiple times.
3. **Read/Write (RW):** Data can be updated or erased multiple times.

Memory Structure

- **Total Memory:** 1024 bytes (1 KB)
- **Total Sectors:** 16
- **Blocks per Sector:** 4
- **Total Blocks:** 16 × 4 = **64 blocks**
- **Each Block Size:** 16 bytes

1 Sector
+----------------------+
| Block 0 (16 Bytes)|
+----------------------+
| Block 1 (16 Bytes)|
+----------------------+
| Block 2 (16 Bytes)|
+----------------------+
| Block 3 (16 Bytes)| ← Sector Trailer (Keys & Access Bits)
+----------------------+
This structure is repeated for all **16 sectors**.

Storage Capacity
Each block stores **16 bytes**.

- **1 byte = 1 ASCII character**
- Therefore, **1 block can store up to 16 characters**. This block stores **16 ASCII characters**.

A **MIFARE Classic 1K RFID tag** contains:

Note: In **Sector 0**, **Block 0** stores the UID (Unique Identifier) and manufacturer data. It is not used for storing user data.

RFID connection to Raspberry Pi 4

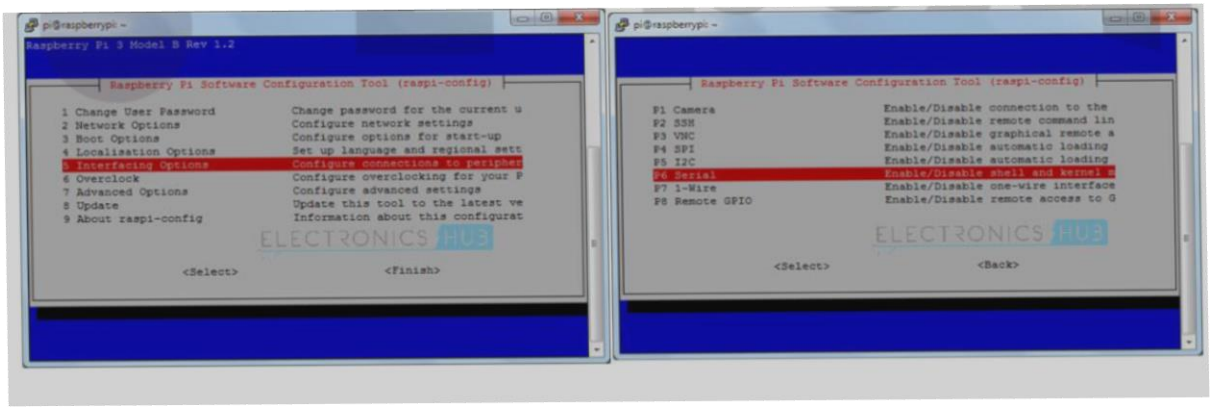
Module	RPi	Phy Pin
1) Vcc	3.3V	1
2) Gnd	Gnd	6
3) SDA	CS0 (SPI0)	24
4) SCK	SCLK	23
5) MOSI	MOSI (0)	19
6) MISO	MISO (0)	21

Commands -

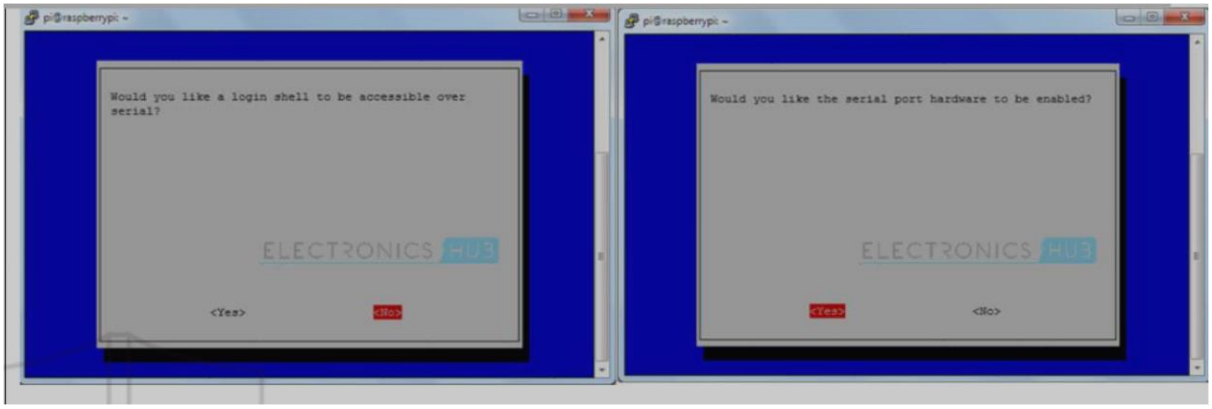
```
sudo raspi-config                    #(Enable SPI)
lsmod | grep spi
sudo apt-get update
sudo apt-get upgrade
sudo pip3 install mfrc522
```

sudo raspi-config

In the “Interfacing Options”, select the “Serial” option.



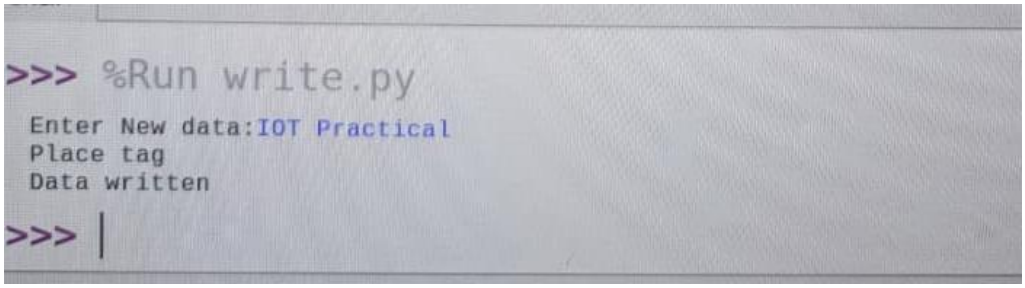
Now, it asks whether you would like to access the login shell over serial communication. Select “No” option. Then, it asks do you want to enable the Serial Port Hardware. Select “Yes” option.



Source Code -

```
#To write data
import RPi.GPIO as GPIO
from mfrc522 import SimpleMFRC522
reader = SimpleMFRC522()
try:
    text = input (“Enter New data:”)
    print('Place tag')
    reader.write (text)
    print ("Data written successfully!")
finally:
    GPIO. cleanup()
```

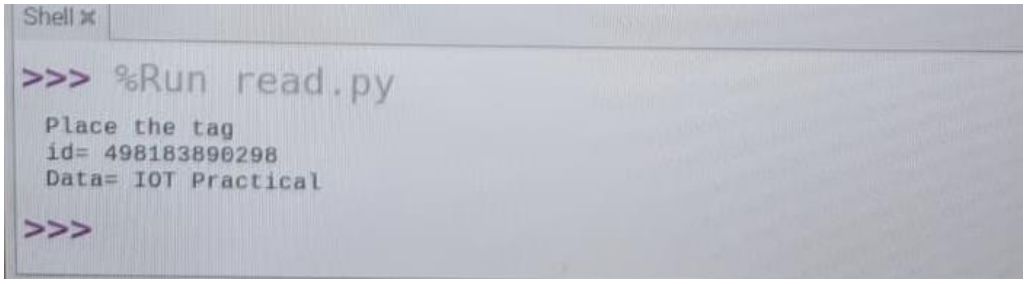
O/P –




```
#To read data :
import RPi.GPIO as GPIO
from mfrc522 import SimpleMFRC522
reader = SimpleMFRC522()
try:
    print("Place the RFID tag near the reader...")
    id, text = reader.read()
    print("Tag ID:"  ,id)
```

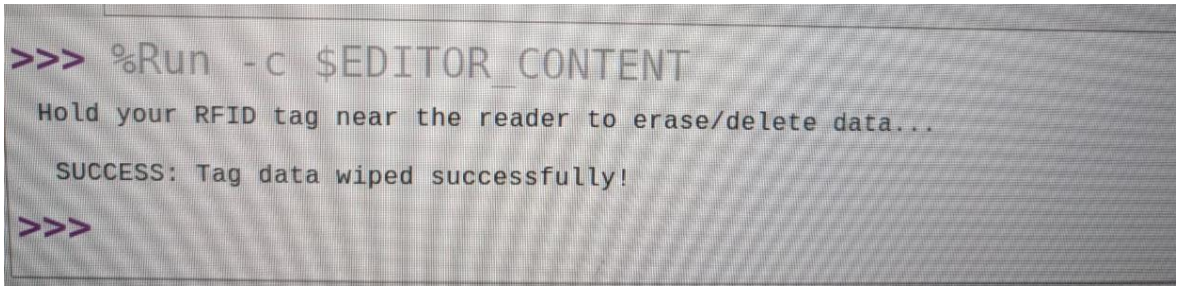
```
    print("Data stored:"  ,text)
finally:
    GPIO.cleanup()
```

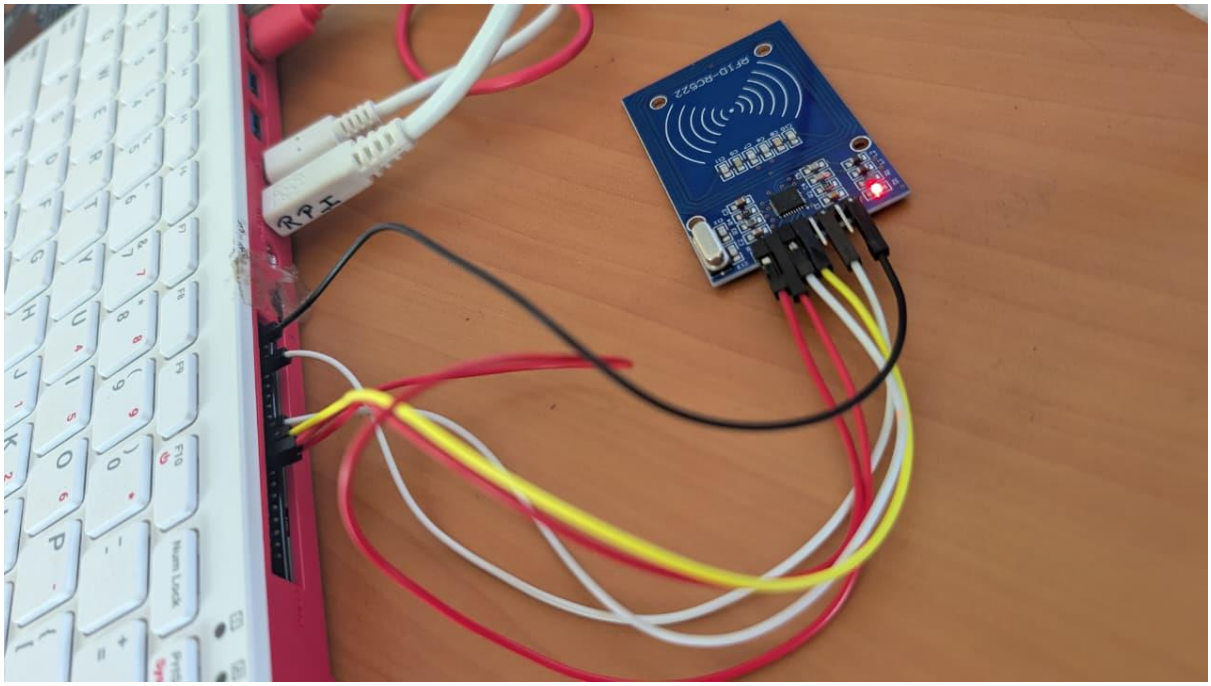
O/P –




```
#To erase data
import RPi.GPIO as GPIO
from mfrc522 import SimpleMFRC522
reader = SimpleMFRC522()
try:
    print("Hold your RFID tag near the reader to erase/delete data...")
    # Overwrite the tag data block with empty text (blank spaces)
    reader.write("                ")  # 16 spaces (1 block size on MIFARE 1k)/ (" " *
16)
    print("SUCCESS: Tag data wiped successfully!")
except Exception as e:
    print(f"An error occurred: {e}")
finally:
    GPIO.cleanup()
```

O/P –




RPI